Koustava Ramesha

Software Engineer | Full Stack & Cloud-Native Development

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Summary

Software Engineer with production experience building and deploying cloud-hosted web applications using **React**, **Next.js**, **Node.js**, **Firebase**, and **Supabase**. Experienced in designing **RESTful APIs**, authentication systems, and real-time services, with a focus on scalability, reliability, and performance. Comfortable across the stack, from backend services and data modeling to **containerized deployments** and **Git-based CI-style workflows**.

Technical Skills

- **Languages:** TypeScript, JavaScript, Python
- **Backend & APIs:** Node.js, Express.js, RESTful APIs, JSON, WebRTC Signaling, Client-Server Architecture
- **Cloud & DevOps:** Firebase, Firestore, Supabase, Docker, Vercel, Git, GitHub, Agile/Sprint Workflows
- **Databases:** Firestore, MongoDB, MySQL, SQLite
- **Architecture & Engineering:** System Design, Microservices, OOP, Debugging, Performance Optimization
- **Frontend:** React.js, Next.js, JSX, Responsive UI Design, Component-Based Architecture
- **Additional:** Google Genkit, Gemini/LLM, YOLO, OpenCV, OCR

Experience & Projects

STR Mix – Admin Dashboard
Next.js 15, React 19, TypeScript, Firebase, Firestore Live | Codebase

- Developed and deployed a production web application serving **1,000+ monthly client visits**, building end-to-end workflows for content management, portfolio administration, and customer reviews.
- Integrated **Firebase Authentication** and real-time **Firestore** data synchronization to support authenticated administrative operations and dynamic content management, maintaining **99.9% system uptime**.
- Optimized application routing, rendering, and data fetching, reducing page-load time by **35%** and achieving a **95+ Lighthouse Performance Score**.
- Developed reusable TypeScript components and streamlined backend data flows, reducing administrative response times by **40%** across content and review-management workflows.

Aether – Decentralized P2P Chat Application
React, TypeScript, Node.js, WebRTC Live | Codebase

- Architected a modular **React and TypeScript** client with a custom **Node.js signaling backend**, enabling real-time peer-to-peer communication using WebRTC.
- Engineered a signaling service supporting **100+ concurrent active connection channels**, handling peer discovery, connection establishment, and signaling message exchange.
- Implemented client-side **AES-256 encryption** and wallet-signature verification to protect message handling and verify communication participants.

Industrial Safety AI Surveillance System
Python, PySide6, YOLO, ByteTrack, OpenCV, SQLite Codebase

- Architected a multi-camera surveillance application using **Python, PySide6, and YOLO**, integrating real-time object detection and ByteTrack-based tracking for industrial safety monitoring.
- Designed a multi-threaded processing architecture with shared inference management and per-camera worker queues to support concurrent video streams.
- Implemented resilient camera reconnection with exponential backoff, detection stabilization, SQLite-based incident logging, evidence capture, and automated safety alert workflows.

Achievements

- **Top 10 Finalist**, Hack-to-Hire National Level Hackathon (TRIKAYA × BeetleX) – among **4,000+ participating teams nationwide**.

Education

RV Institute of Technology and Management (RVITM), Bangalore
B.E. – Information Science and Engineering Expected Graduation: 2027